Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Practical – X

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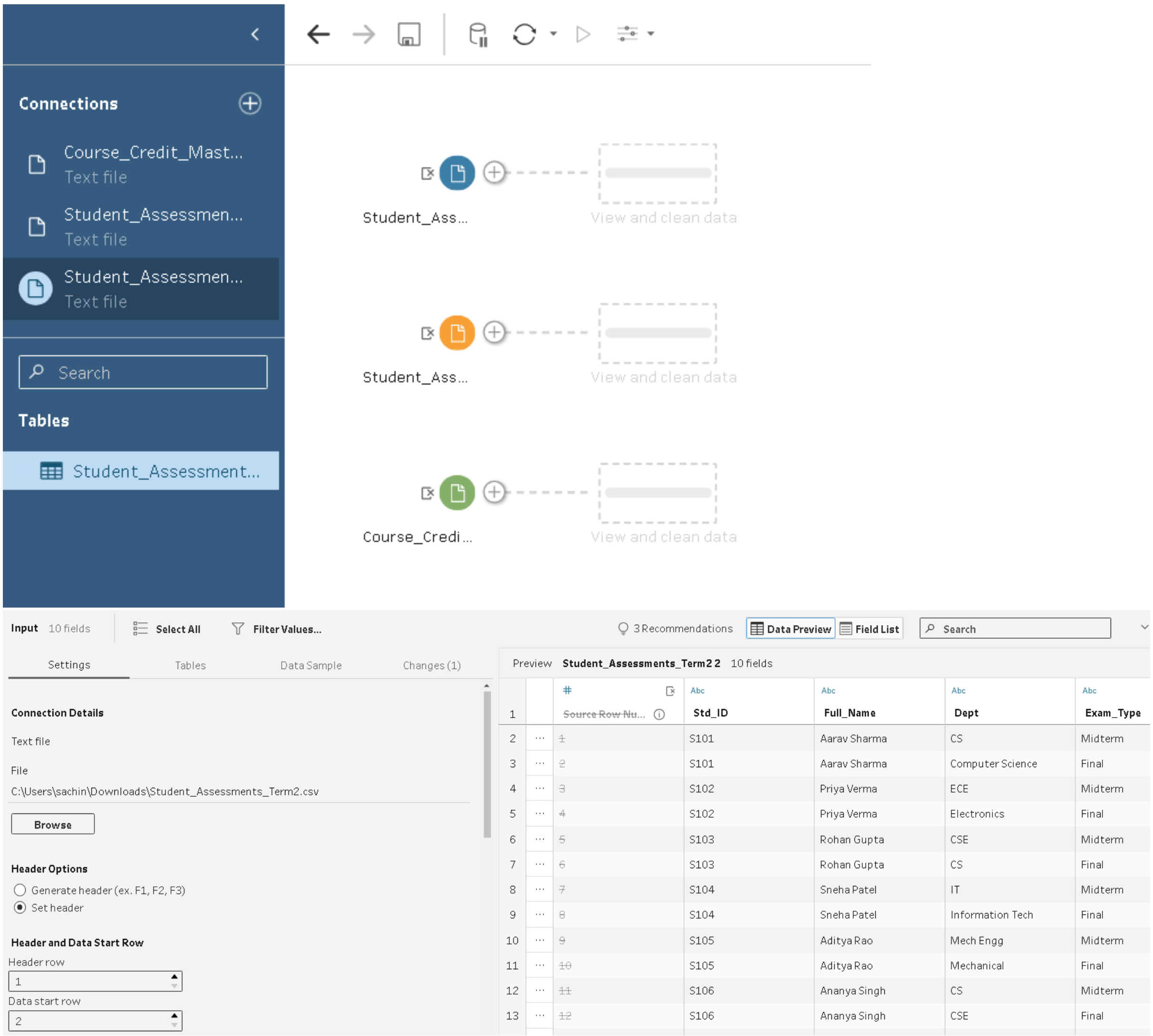
Class: TYIT	Batch: 1
Date of Assignment: 03/09/2026	Date/Time of Submission: 03/09/2026

Data Preparation and Transformation using Tableau Prep

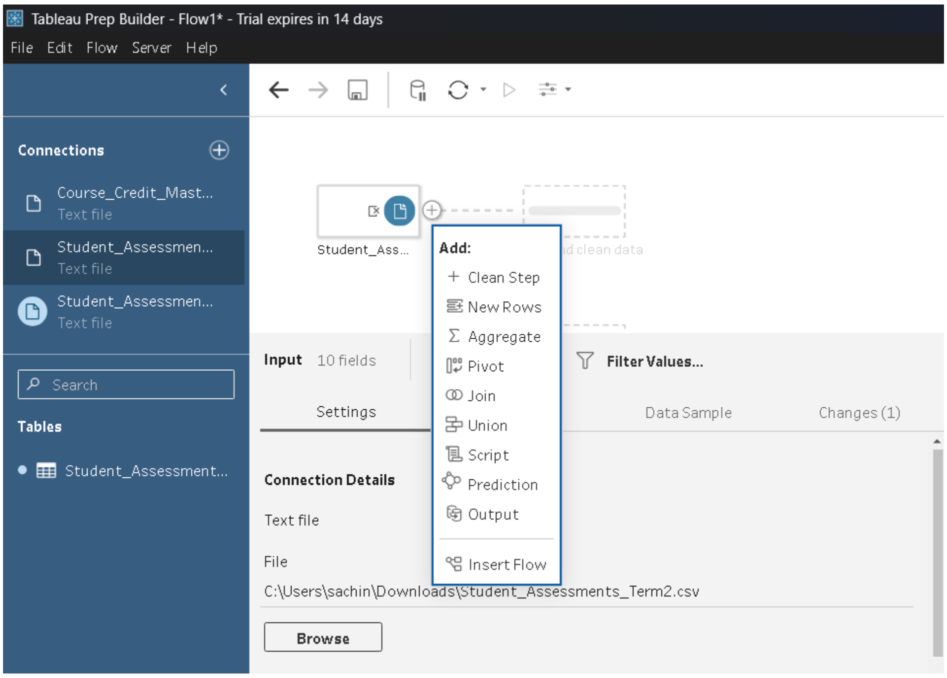
a) Connect and load data into Tableau Prep

Step 1: Select connect to data and click on the text file to upload all three file and the data will be loaded and shown.

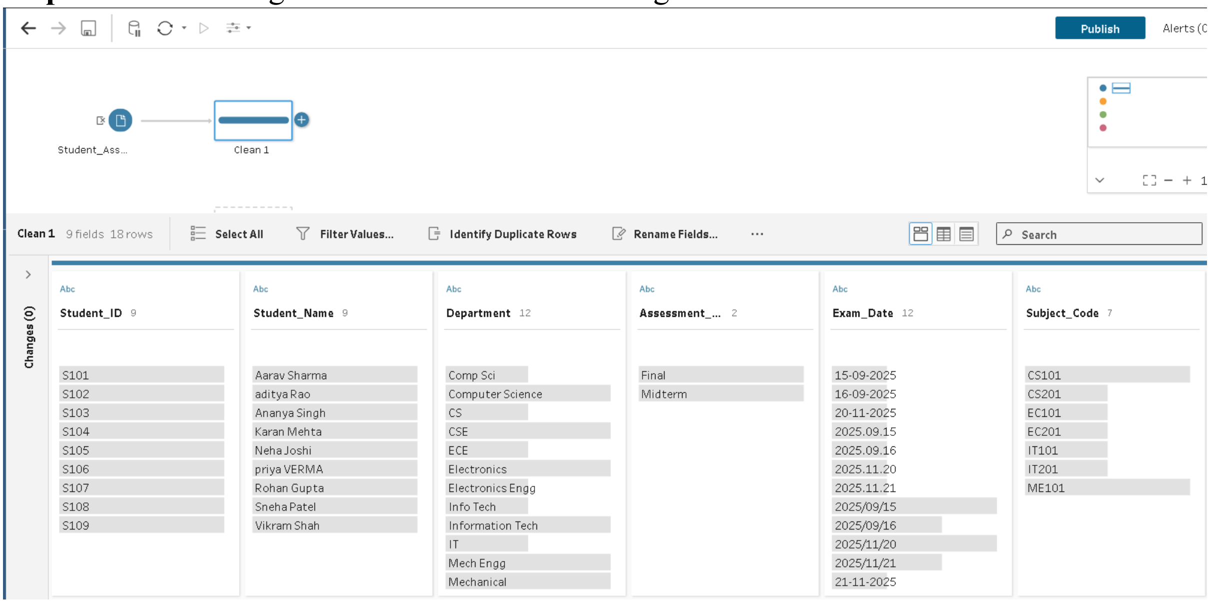


b) Clean and profile data

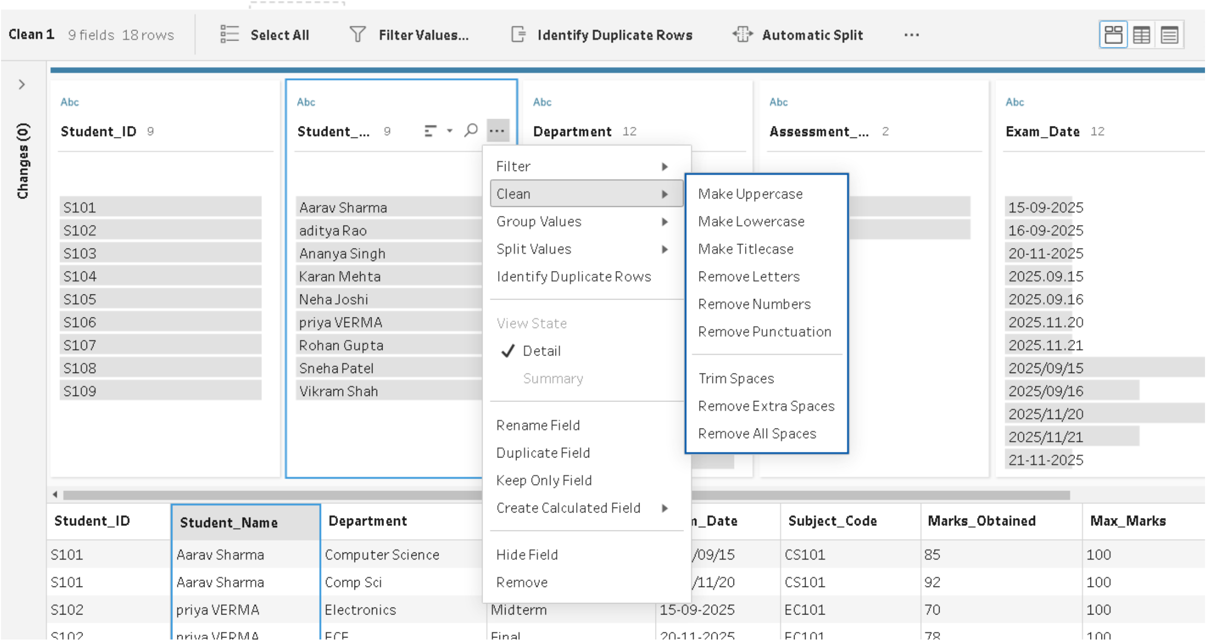
Step 1: Click on the Clean steps.



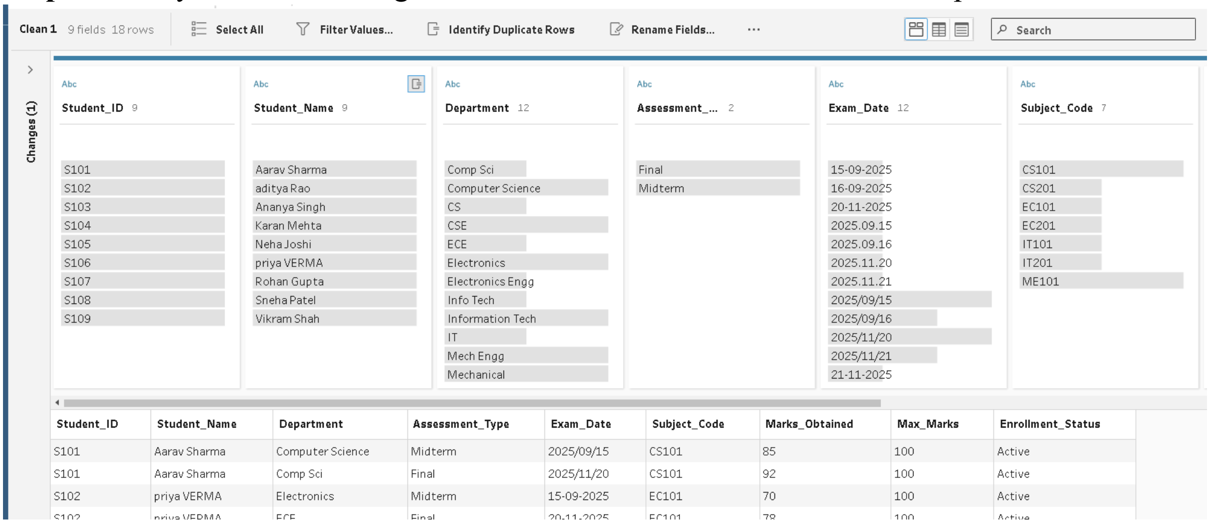
Step 2: After clicking on it the data will have changes shown below.



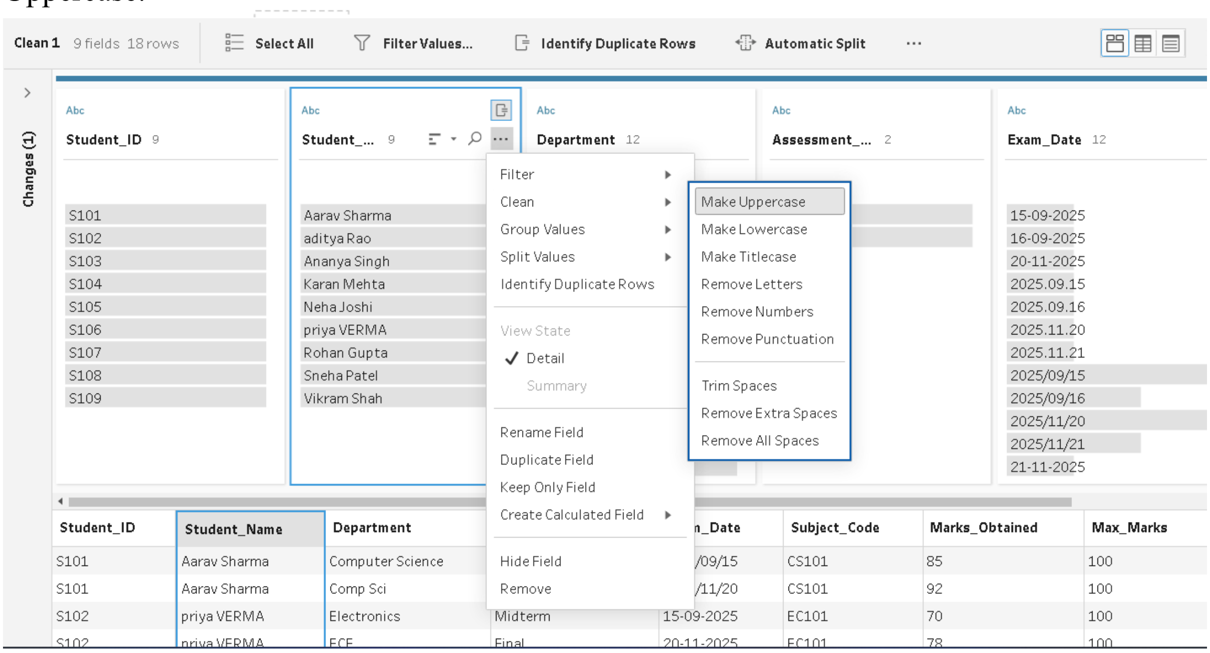
Step 3: After changes we need to see the Student_Name card and click on … we will click on clean →Trim Spaces



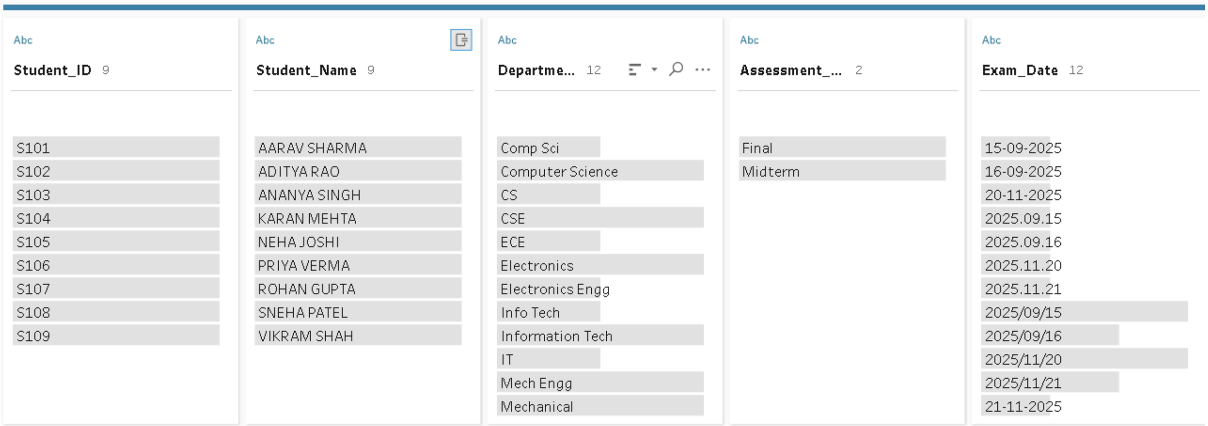
Step 4: Now you can see changes in data it removes and trims unwanted spaces



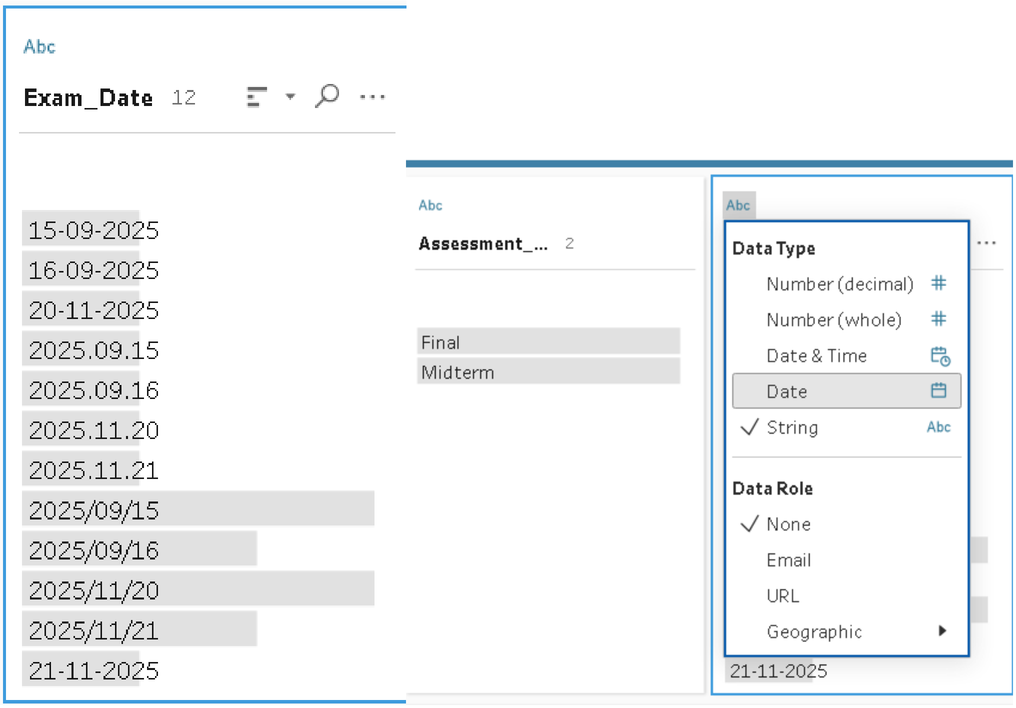
Step 5: Now again go on Student_Name card and click on … we will click on clean →Make Uppercase.



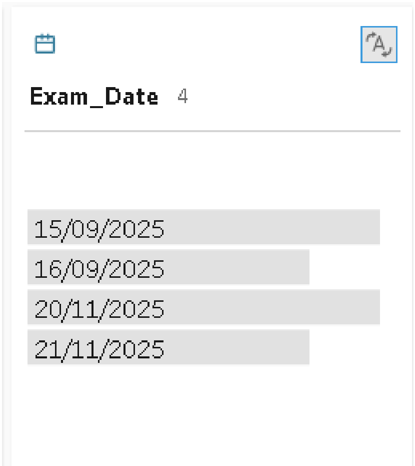
Step 6: See changes in data



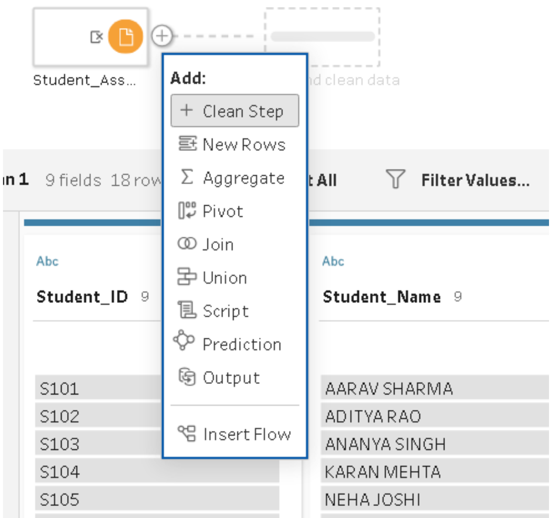
Step 7: Now click on Abc shown in Exam_Date change it into Date



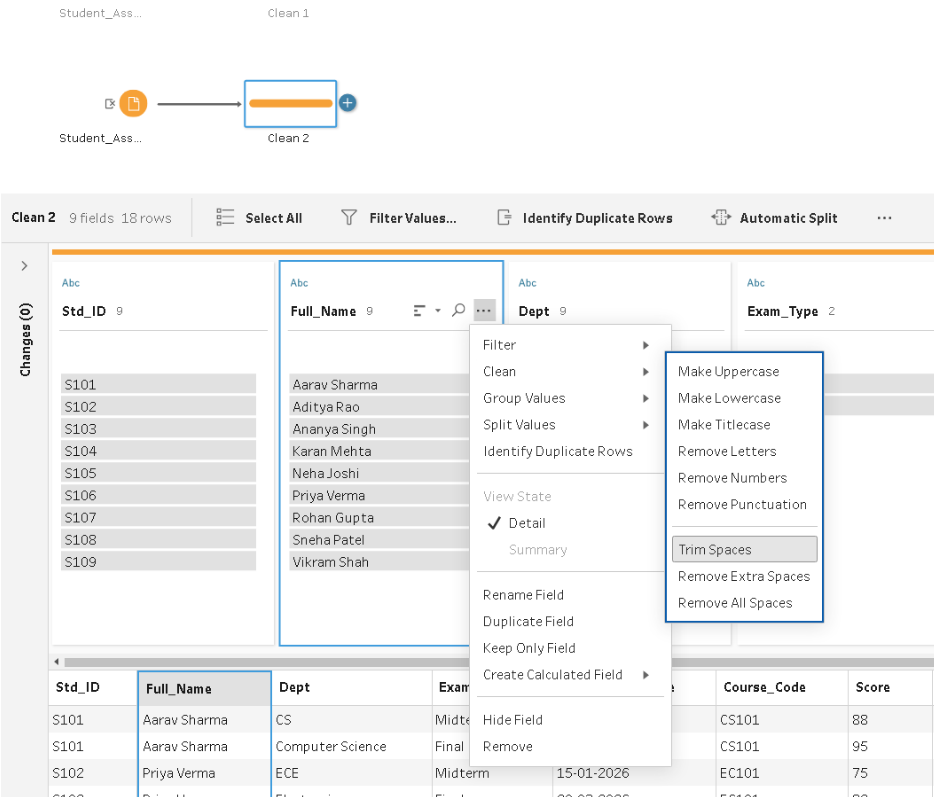
Step 8: Now see the changes



Step 9: Now open Student_Assessments_Term2 and Clean Step

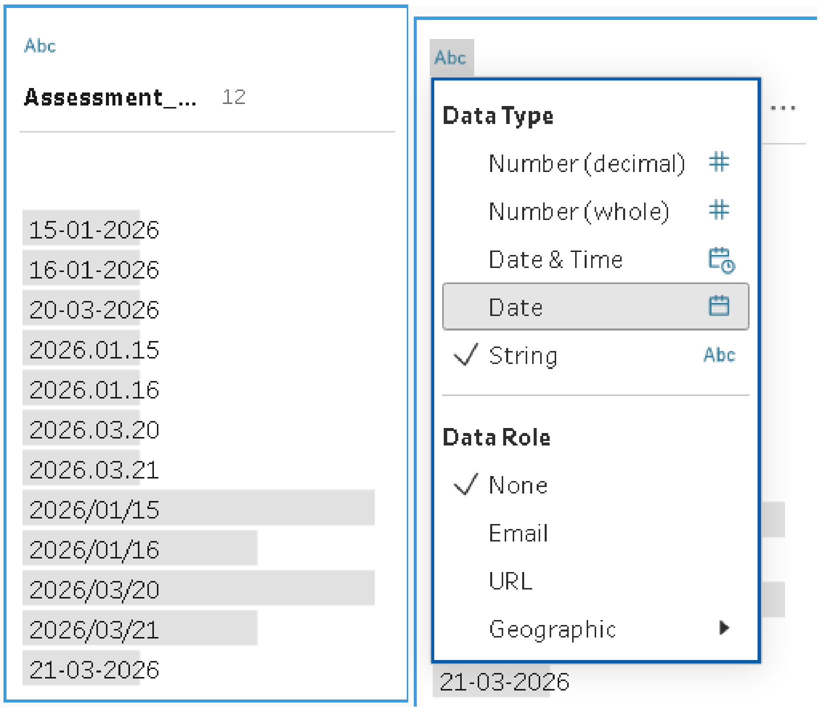


Step 10: Now do same step as click on three dot of Full_Name and click on Trim Spaces

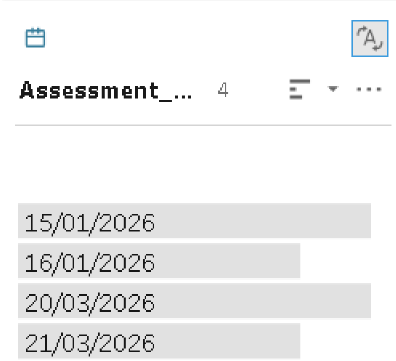


Step 11: See changes in Full_Name.

Step 12: On Assessment_Date click on abc and change the data type to Date

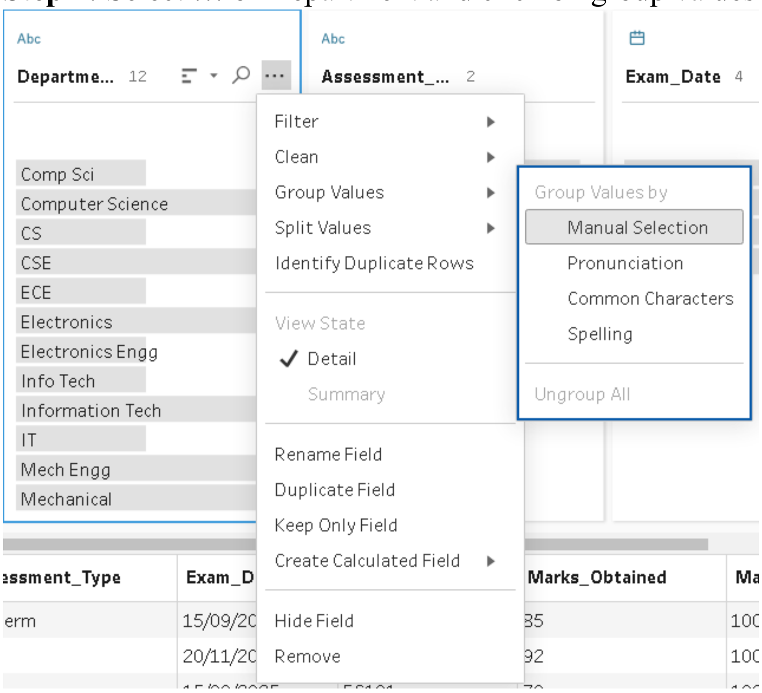


Step 13: See changes in Assessment_Date

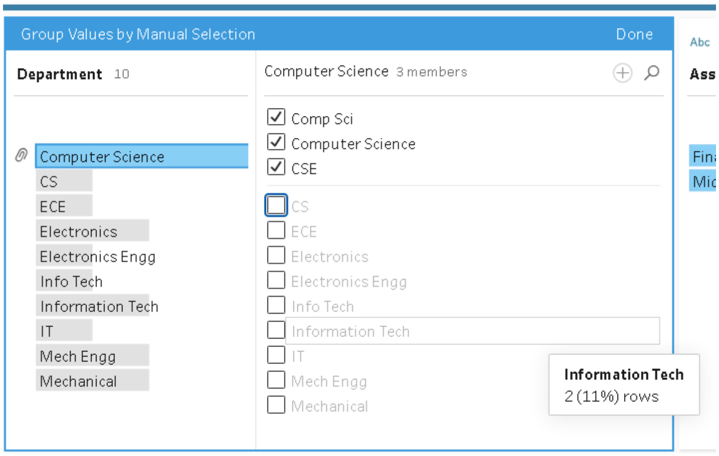


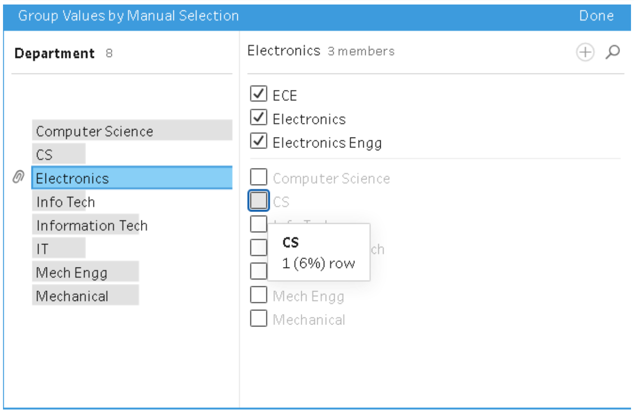
c) Group and standardize values

Step 1: Select … of Department and click on group values → Manual Selection

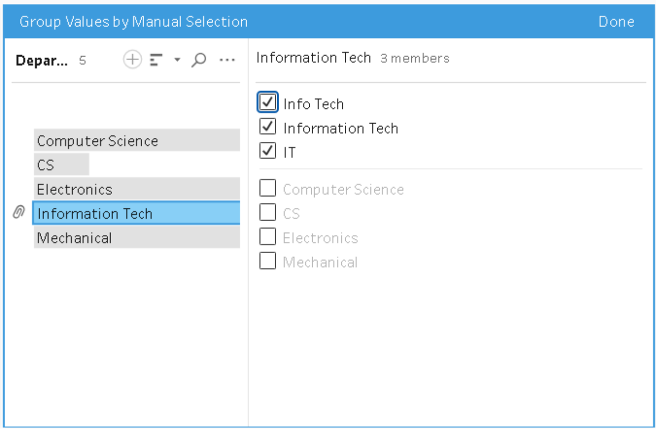


Step 2: Select manually all details you need in computer science and Electronics

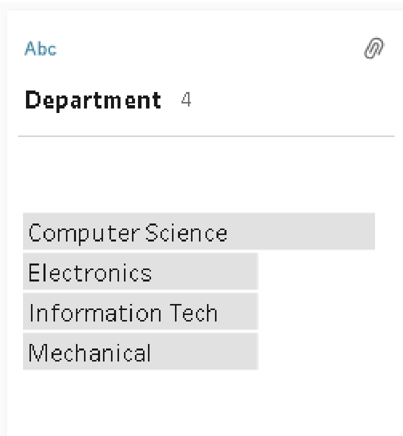




Step 3: Now select other as shown below in Information Tech and for mechanical as well

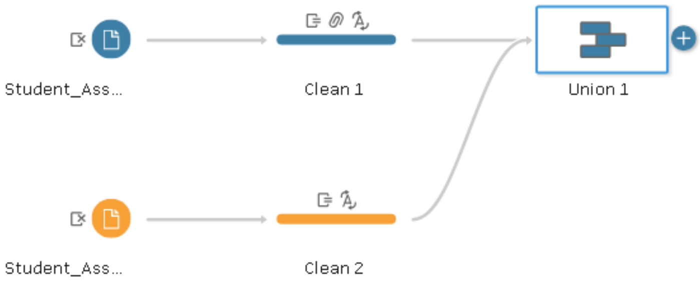


Step 4: Now see the changes in department showing 4 columns named Computer Science, Electronics, Information tech, Mechanical.

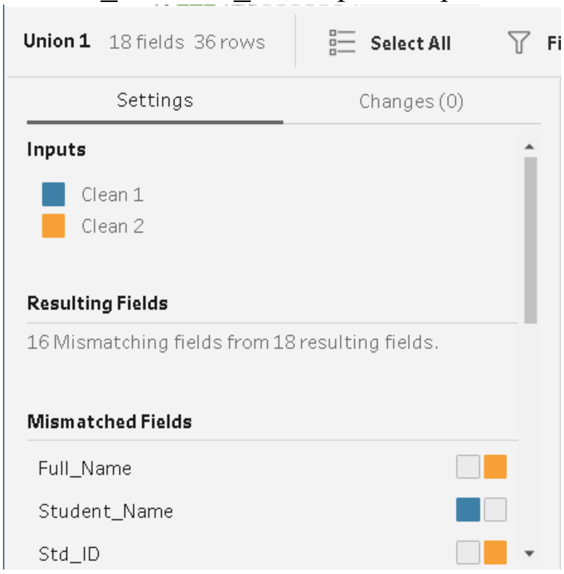


d) Merge mismatched fields

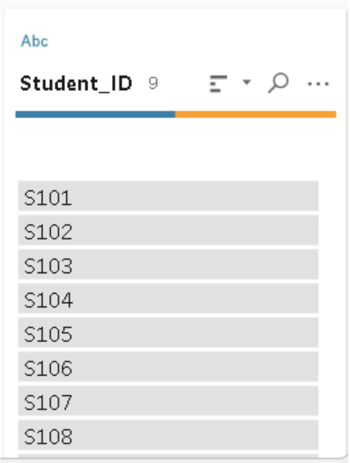
Step 1: Create a union by moving Clean 2 to Clean 1

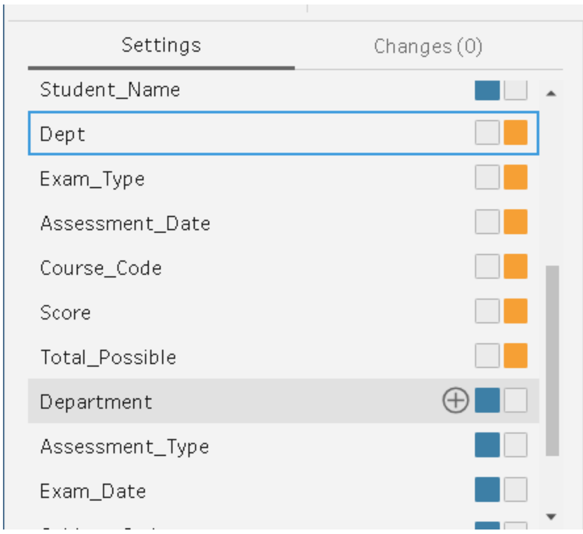


Step 2: Below you saw the mismatched fields you need to drag and add the same like Student_ID to Std_ID, Dept to Department

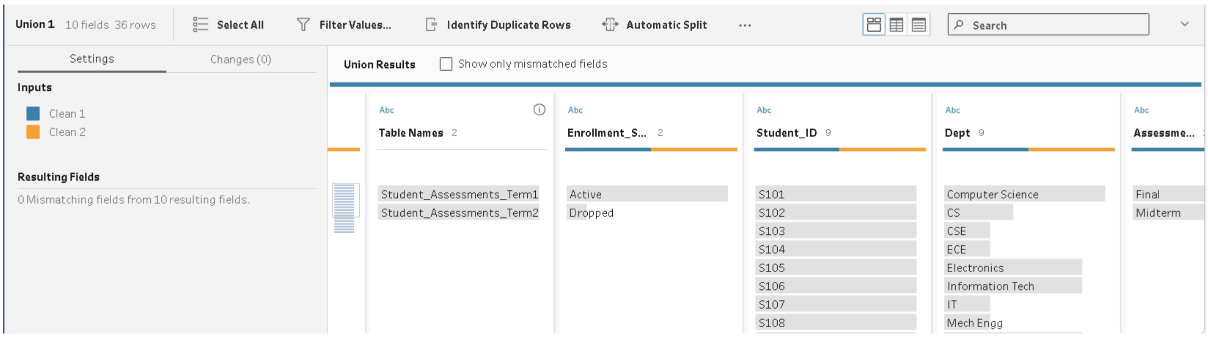


Step 3: You will saw the merged values of both Student_ID and Std_ID do the same for others as well



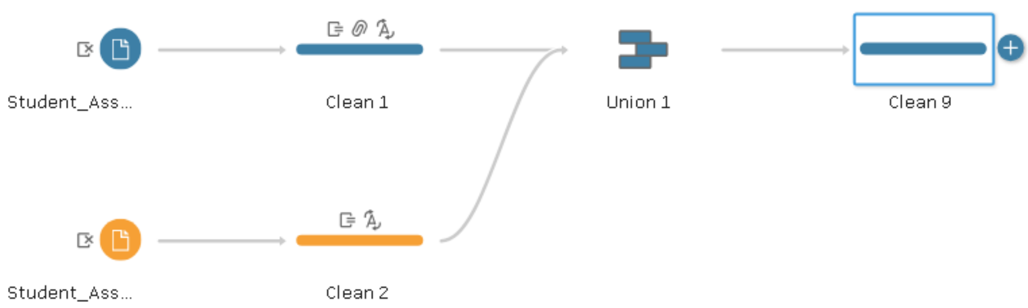


Step 4: Now click and do for all other mismatched fileds and you will see all done as shown below

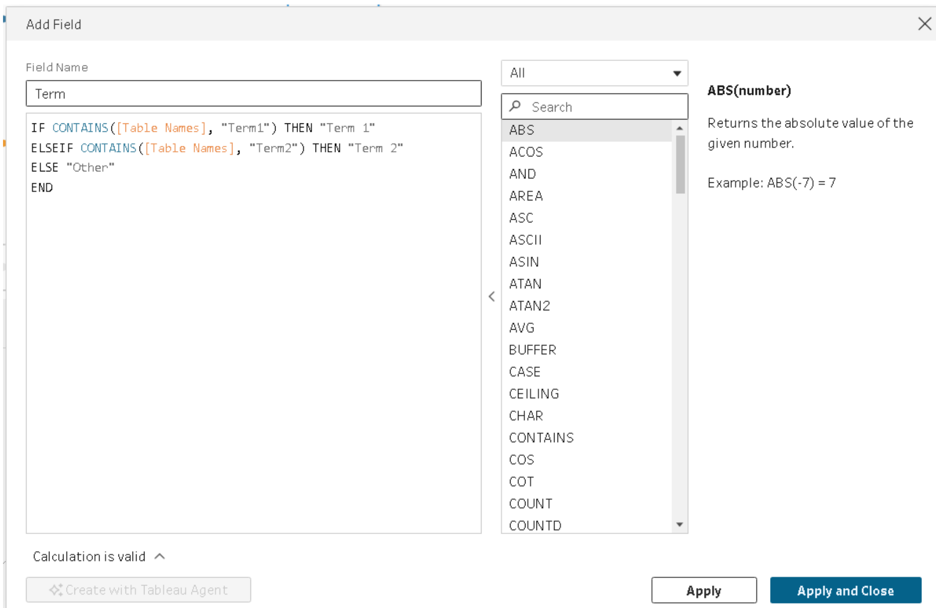


e) **Perform aggregations in Prep**

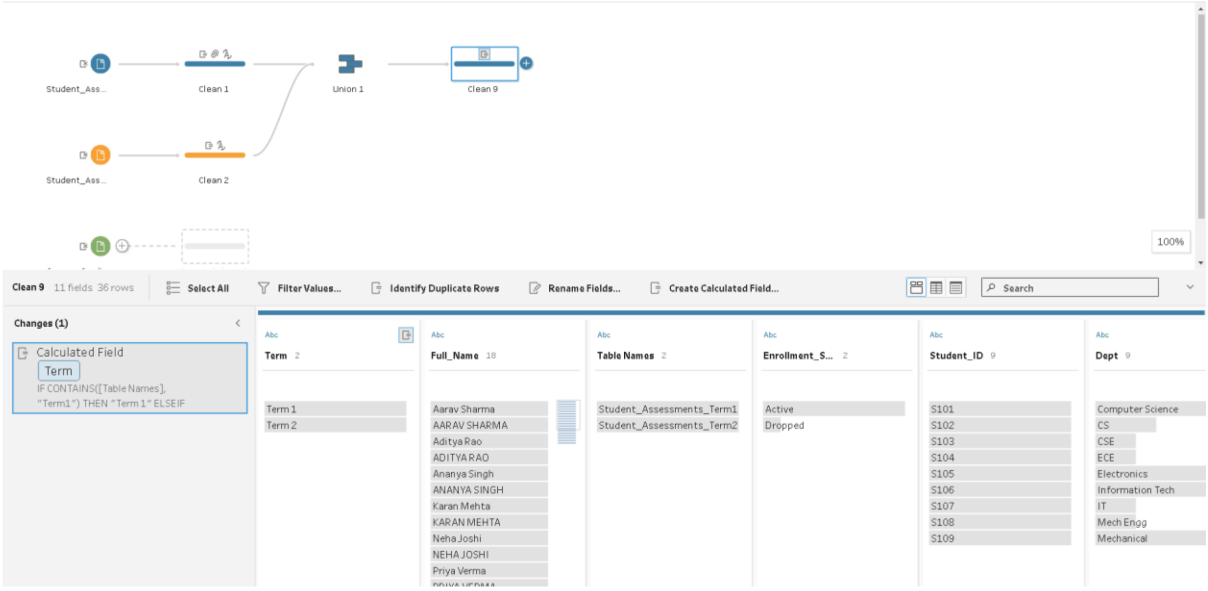
Step 1: Add clean step for Union 1.



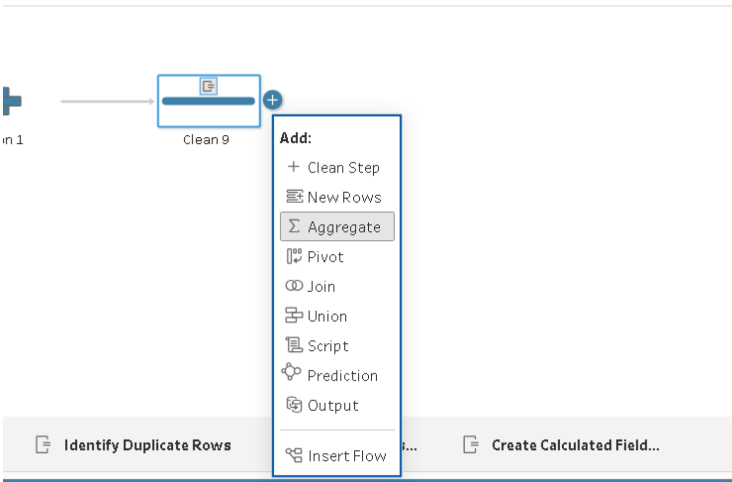
Step 2: Above the table you will see Create Calculated field and fill the details shown below as Field Name and Formula then Apply and Close.

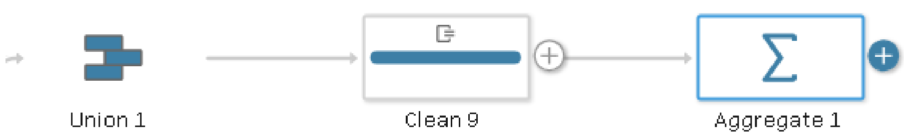


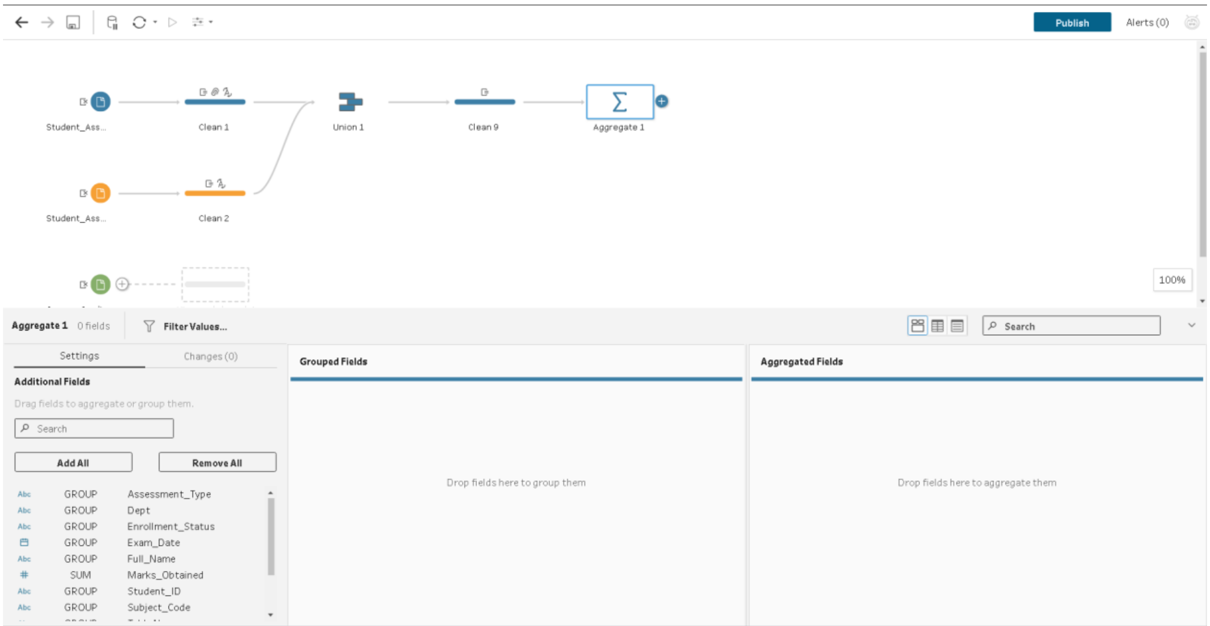
Step 3: You will see the output after changes shown below



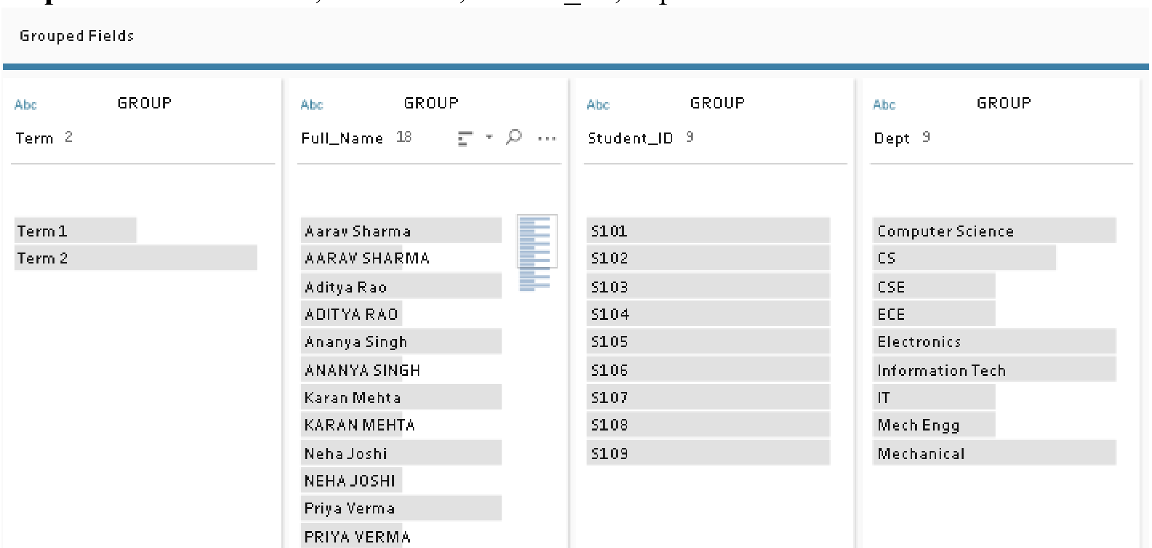
Step 4: Select Clean 9 and add Aggregate .



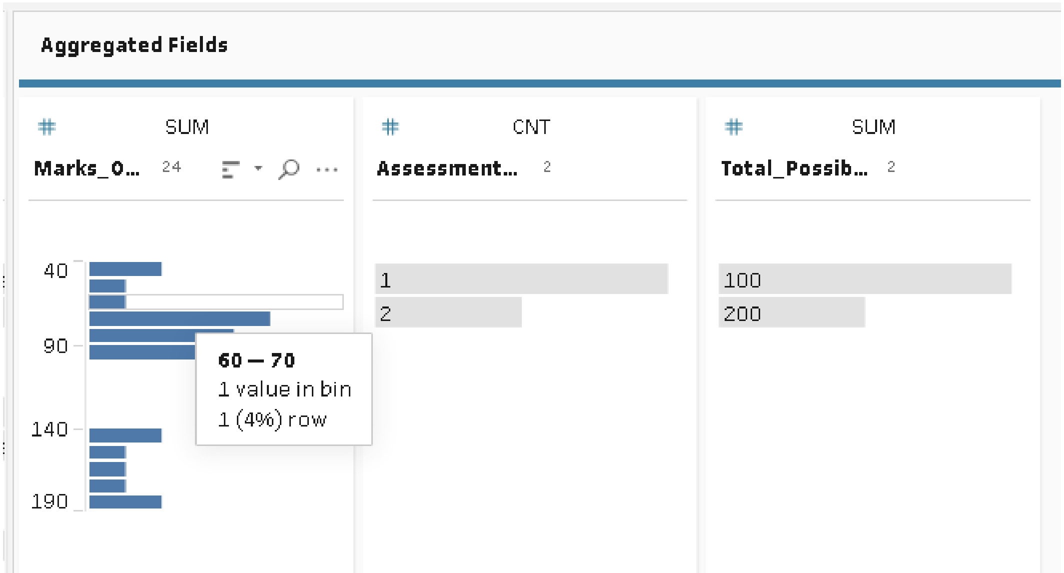




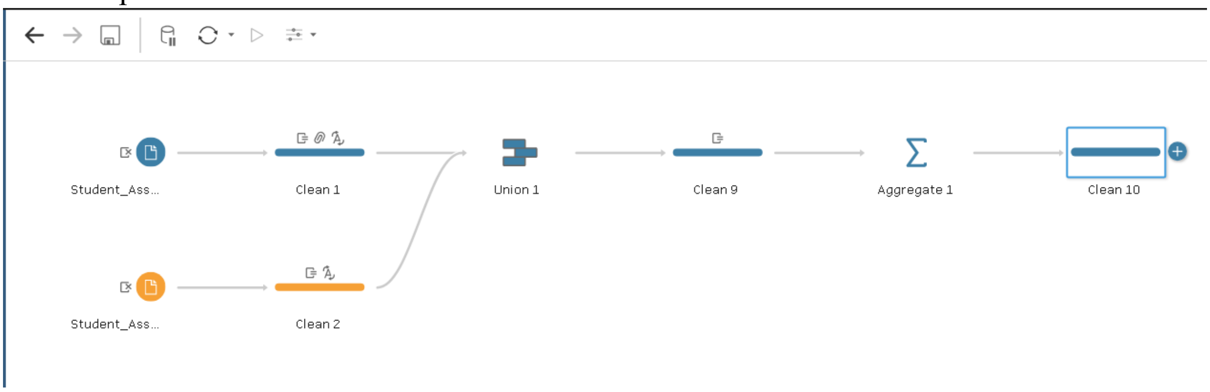
Step 5: Now addd Term,Full Name,Student_ID,Dept



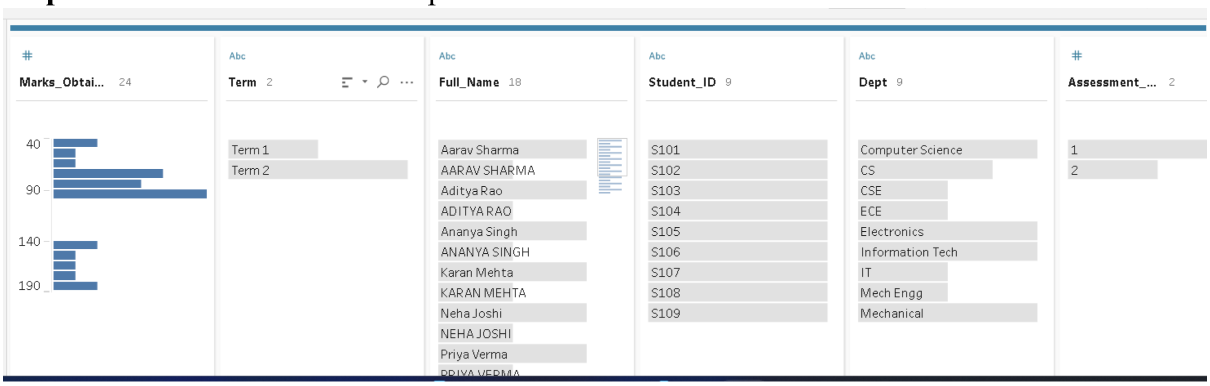
Step 6: Add the other three in Aggregate Fields as well.



Step 7: Now in Aggregate 1 we need to click on Clean Step and the Clean 10 will created open it

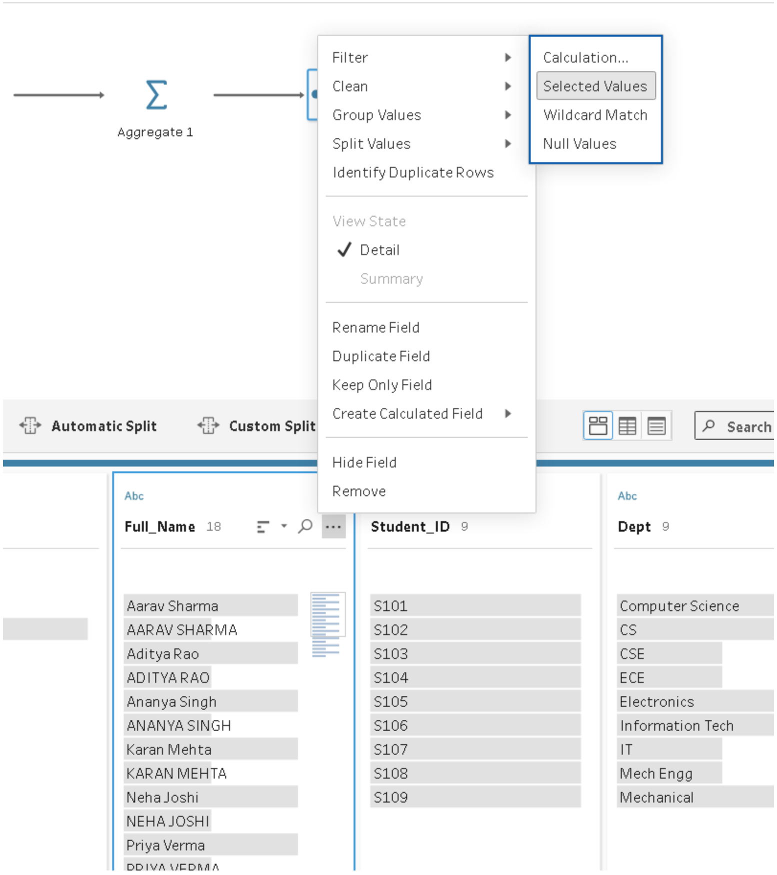


Step 8: Below show the final output.

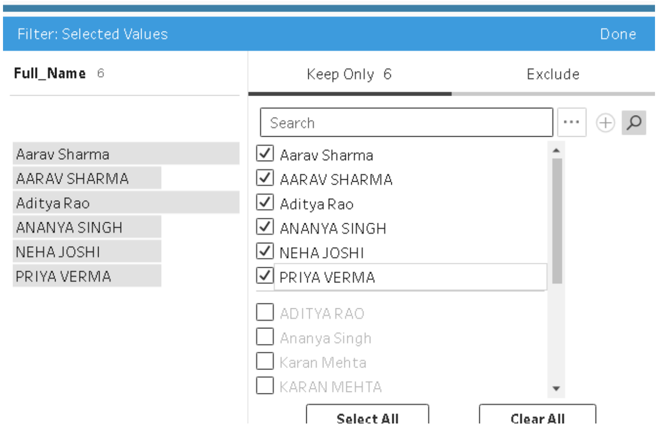


f) Apply filters and transformations

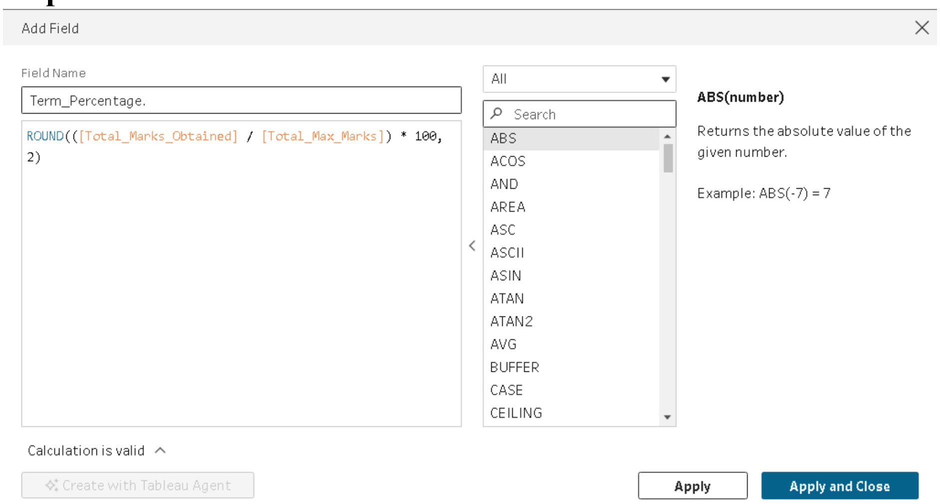
Step 1: Click on the three dot of Full_Name then Filter→Selected values.



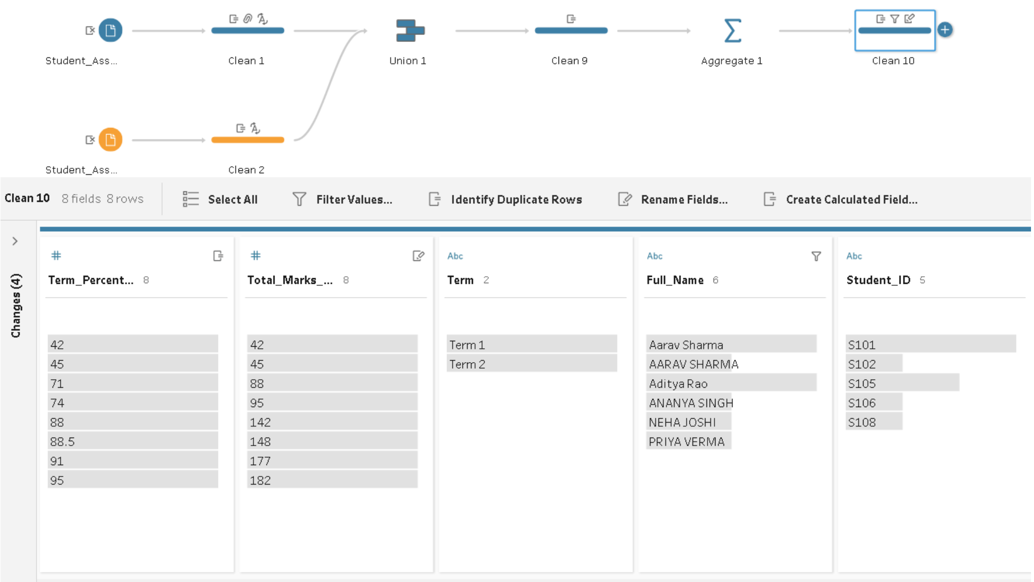
Step 2: Filter the inactive/Dropped Student record and done

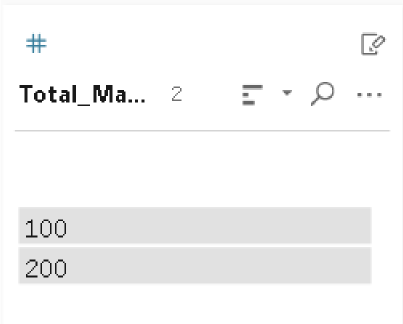


Step 3: Write the filed name and formula shown below

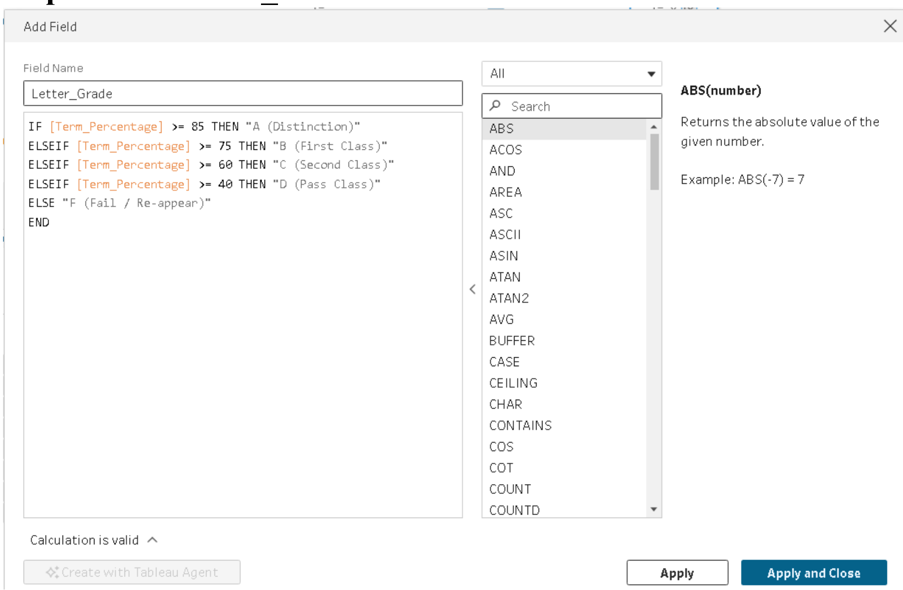


Step 4: See the changes below

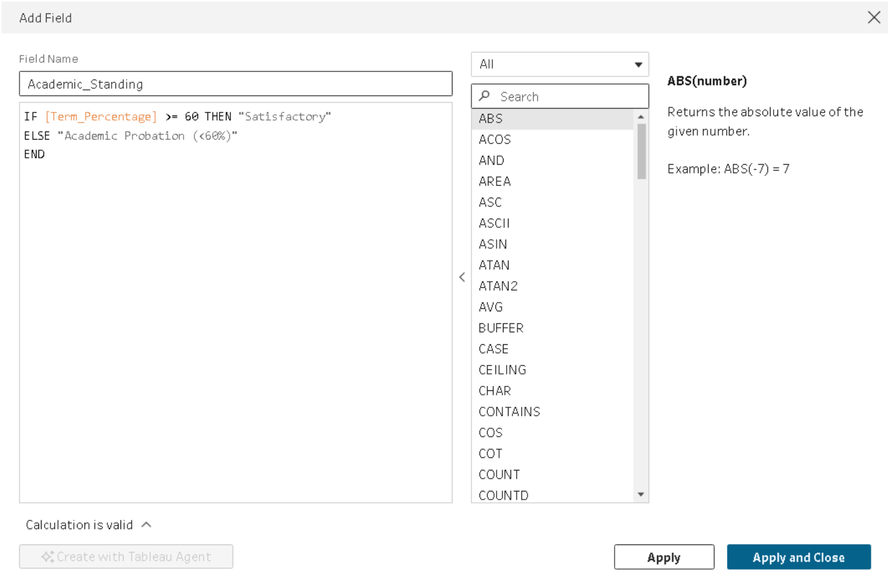




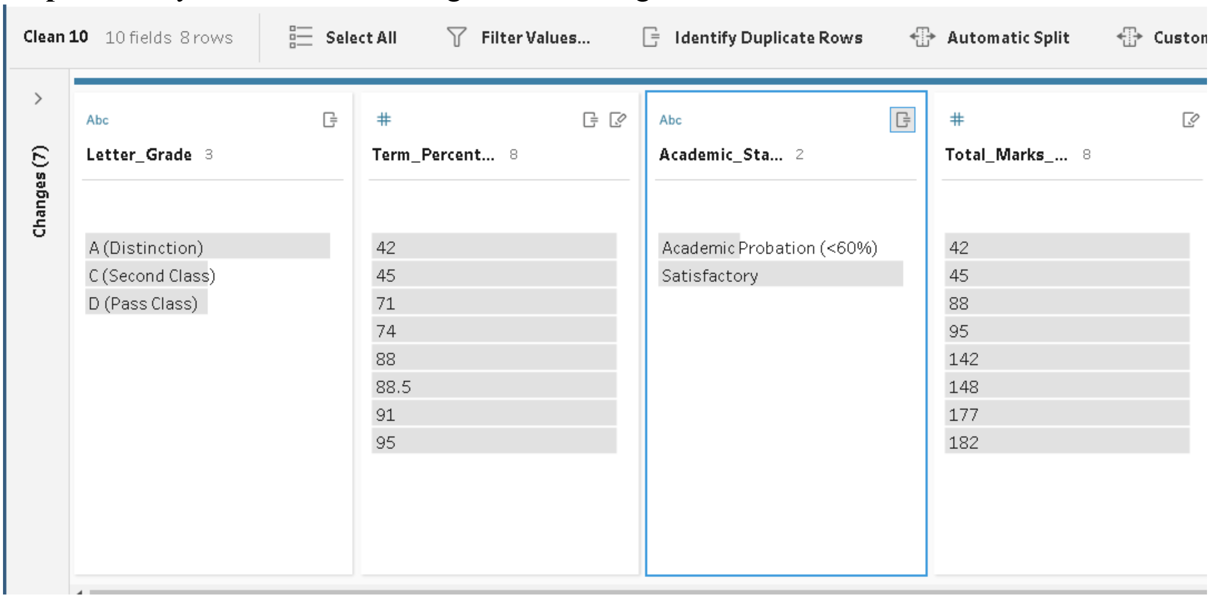
Step 5: Add Letter_Grade and its formula



Step 6: Add Academic_Standing and formula as shown below

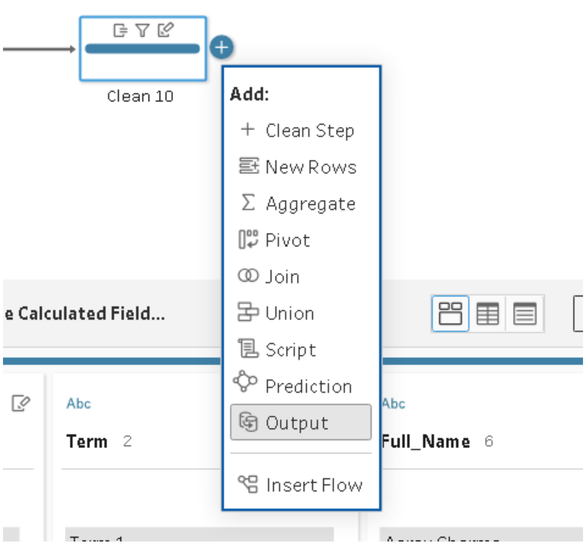


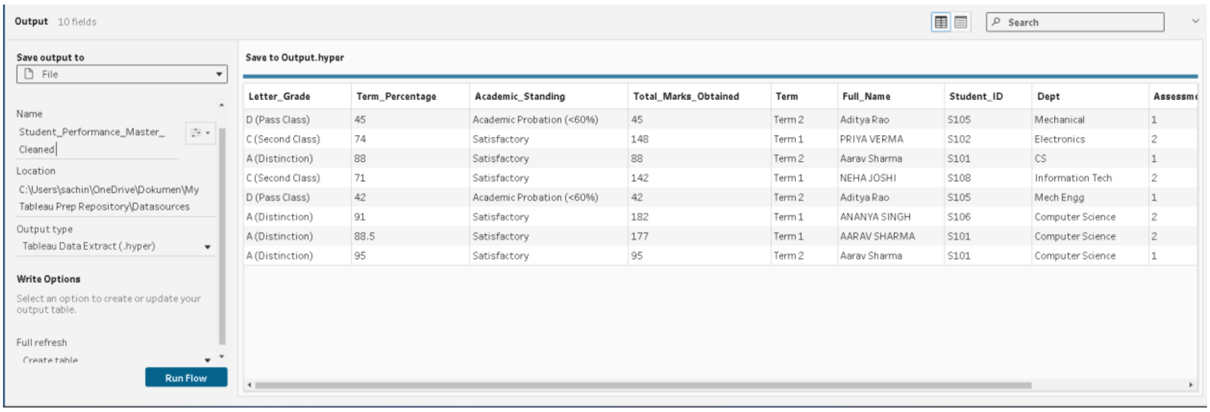
Step 7: Now you will see the changes after adding the above formulas



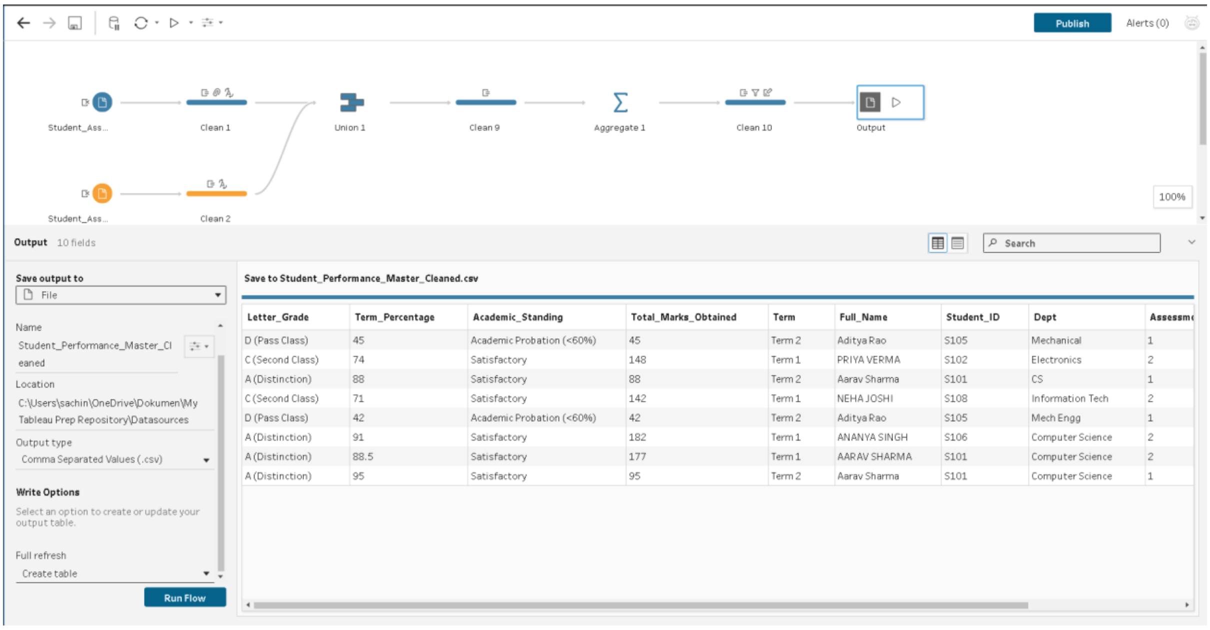
g) Create and automate data preparation flows

Step 1: Click on the clean 10 and click on + option and click on output

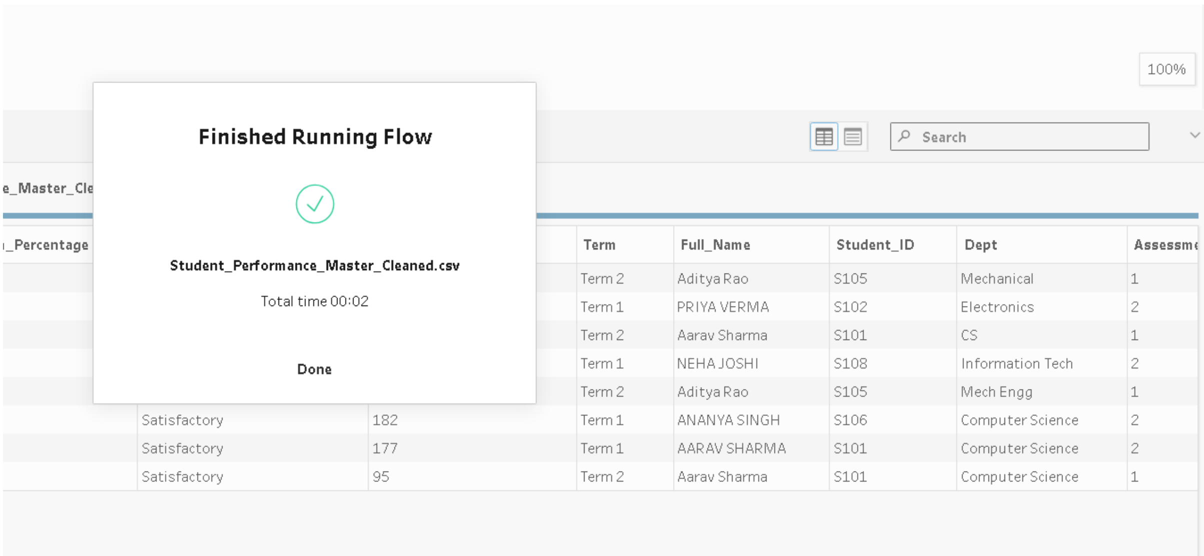




Step 2: Change the name output into Student_Performance_Master_Cleaned and output type into csv



Step 3: Now you need to click on Run Flow and then the output is shown below



Step 4: Write the below command in cmd run as administrator

```
C:\Windows\System32>dir "C:\Users\sachin\OneDrive\Dokumen\My Tableau Prep Repository\*.tfl" /s
 Volume in drive C is Windows-SSD
 Volume Serial Number is D4E0-1C17

 Directory of C:\Users\sachin\OneDrive\Dokumen\My Tableau Prep Repository\Flows

04-09-2026  12:43            28,734 Practical 10.tfl
04-09-2026  12:43            28,734 Student_Performance_Prep_Flow.tfl
               2 File(s)         57,468 bytes

     Total Files Listed:
               2 File(s)         57,468 bytes
               0 Dir(s)  189,394,382,848 bytes free
```

```
C:\Windows\System32>"C:\Program Files\Tableau\Tableau Prep Builder 2026.2\scripts\tableau-prep-cli.bat" -t "C:\Users\sachin\OneDrive\Dokumen\My Tableau Prep Repository\Flows\Practical 10.tfl
JAVA_HOME is set to : C:\Program Files\Tableau\Tableau Prep Builder 2026.2\scripts\..\Plugins\jre temporarily
The connection file is in V1 format.
Preparing to run the flow : C:\Users\sachin\OneDrive\Dokumen\My Tableau Prep Repository\Flows\Practical 10.tfl
Loading the flow.
Creating temp directory at C:\Users\sachin\AppData\Local\Temp\prep-cli-15998479297145271073
Product license verified.
Loaded the flow.
Checking the flow document for errors.
Flow Document has no errors.
Preparing to execute the flow.
Flow Execution Status: Running
Flow Execution Status: Finished
Finished running the flow successfully.
C:\Windows\System32>
```

Step 5: Now open the file in tableau desktop and add the column and rows as shown below and you will see the final output.